

李伟凯 学术简历

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教育经历

- 2018 – 2022 / 博士 / 计算机科学与技术 / 南京航空航天大学
- 2015 – 2018 / 硕士 / 计算机技术与应用 / 重庆交通大学
- 2011 – 2015 / 学士 / 信息与计算科学 / 重庆交通大学

科研项目

- 域对齐外的无监督域适应研究及其拓展 / 国家自然科学基金青年项目 / 主持
- Advanced School in Applied Machine Learning / 国家自然科学基金国际合作交流项目 / 主持
- 抑郁症异质性脑网络的多模态神经生理机制解析与数字孪生闭环经颅磁刺激机器人关键技术研究 / 山东省重大基础研究项目 / 主持
- 域对齐外的无监督域适应探索 / 工信部重点实验室开放项目 / 主持
- 基于 PET/MR 多模态脑网络的帕金森症早期诊断及识别 / 分子影像湖北省重点实验室开放项目 / 主持
- 基于多模态影像的自闭症多尺度模式分析与研究 / 重庆市教委科研项目青年基金 / 主持
- 基于多中心迁移学习的脑网络疾病诊断方法研究 / 重庆市自然科学基金面上项目 / 主持
- 调度协议和网络攻击下随机复杂网络的保概率同步性及安全控制研究 / 重庆市科技局自然科学基金创新发展联合基金 / 主研
- 自监督学习本质的有效性探究、论证及拓展增强 / 国家自然科学基金面上项目 / 主研

论文成果

- 2026 / Universum Inspired Graph Contrast Learning for Major Depressive Disorder Identification: In Functional Brain Network View. H Qin, G Hong, D Ma, Z Wang, L Zhao, W Li, X Xu. CNS Neuroscience & Therapeutics 32 (9), e71158, 2026
- 2026 / Multi-Source Multi-View Graph Domain Adaptation with Hyperbolic Residual Encoding for Cross-Site MDD Identification from rs-fMRI. Z Zheng, X Chen, H Jiang, R Tian, Q Cai, J Liu, X Chen, W Li, Y Wang. arXiv preprint arXiv:2607.29531, 2026
- 2026 / Domain generalization meets mutual bi-attention: A novel approach to drug-target prediction. J Yang, W Li, K Qian, Z Cheng. Neurocomputing, 134626, 2026
- 2026 / Enhancing Fatigue Detection Through Heterogeneous Multisource Data Integration and Cross-Domain Modality Imputation. L Cui, Y Wu, Y Tang, W Li. IEEE Transactions on Computational Social Systems, 2026
- 2026 / Hierarchical neighbor integration graph attention network for autism spectrum disorder diagnosis. D Ma, L Peng, L Zhang, W Li, X Gao. Frontiers in Psychiatry 17, 1846996, 2026
- 2026 / TumorAL: Evidence-aware active learning for 3D tumor segmentation. H Wang, J Leng, S Li, Y Zhao, W Li, W Li, X Gao. Neurocomputing, 134239, 2026
- 2026 / Identification of Major Depressive Disorder Using Multiple Functional Connection Patterns. Y Ruan, L Guan, L Peng, L Zhao, Y Huang, W Li, X Gao. CNS Neuroscience & Therapeutics 32 (6), e70951, 2026
- 2026 / Electrocardiographic Alterations Combined with Hematological, Biochemical, and Metabolic Profiles Predict Prognosis in Kawasaki Disease. Q Wang, W Li, J Wan, L Wei, Y Xia, Y Hua, K Zhou, D Qie, W Li, Y Li. Journal of Cardiovascular Development and Disease 13 (6), 228, 2026
- 2026 / Partial domain adaptation enables cross domain cell type annotation between scRNA-seq and snRNA-seq. X Chen, Q Zou, Q Cai, X Chen, W Li, Y Wang. PLOS Computational Biology 22 (5), e1014223, 2026

- 2026 / Bridging the modality reliability gap in drug-target interaction prediction via a confidence-aware multimodal fusion framework. J Yang, J Zhang, K Qian, Q Yang, W Li, Z Cheng. Proceedings of the AAAI Conference on Artificial Intelligence 40 (32), 27529 ..., 2026
- 2026 / Test-time few-shot object detection via dynamic prototype fusion. Y Wu, Y Li, H Wei, W Li, Y Tang. IEEE Transactions on Cybernetics, 2026
- 2026 / Machine learning-based assessments of perioperative features in determining long-term conduct block post-transcatheter closure of ventricular septal defect. Q Wang, Y Xia, L Wei, P Yang, K Zhou, Y Hua, W Li, Y Li. Cardiology in the Young, 1-11, 2026
- 2025 / Attention Transfer Based Hybrid Knowledge Distillation for Multimodal Brain Tumor Segmentation. H Ma, HS Gan, H Mi, P Xu, Y Liu, W Li. 2025 IEEE International Conference on Big Data (BigData), 7503-7510, 2025
- 2025 / More insights into disruption and decoupling of individual metabolic connectomes in Parkinson's disease. D Li, J Yao, W Li, Z Guo, X Lv, L Chen, J Shi, YC Chen, J Ye. Progress in Neuro-Psychopharmacology and Biological Psychiatry, 111578, 2025
- 2025 / Incorporating Negative Node Information in Graph Transformer for MDD Diagnosis. H Song, H Qin, W Li. 2025 International Conference on Cyber-Physical Social Intelligence (CPSI), 1-6, 2025
- 2025 / MVSSL-GT: A Graph Transformer Method with Multi-View Contrastive Self-Supervised Learning. Y Lu, N Tian, W Li. 2025 International Conference on Cyber-Physical Social Intelligence (CPSI), 1-6, 2025
- 2025 / Low-light image enhancement using dual cross attention. Y Ruan, H Ma, D Ma, W Li, X Wang. Engineering Applications of Artificial Intelligence 159, 111501, 2025
- 2025 / Auto-weighted projective one-step multi-view clustering. X Mou, W Li, B Li, J Hu. Neurocomputing, 131960, 2025
- 2025 / Revisiting Multi-Modal Alignment: In Distribution View. W Li, N Tian, Y Li, Y Tang. 2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC) ..., 2025
- 2025 / Quantitative susceptibility mapping shows alterations of brain iron content in children with autism spectrum disorder: a whole-brain analysis. X Xu, Y Li, H Lan, N Ding, W Li, G Zheng, X Song. BMC psychiatry 25 (1), 826, 2025
- 2025 / Cross-modal medical image generation from MRI to PET using robust generative adversarial network. Y Yang, B Li, W Cao, X Chen, W Li. Expert Systems with Applications, 129287, 2025
- 2025 / Leveraging multi-source and heterogeneous signals for fatigue detection. L Cui, Y Wu, T Ying, W Li. arXiv e-prints, arXiv: 2507.16859, 2025
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- 2025 / Timestamp calibration for time-series single cell rna-seq expression data. X Chen, S Lin, X Chen, W Li, Y Li. Journal of Molecular Biology 437 (9), 169021, 2025
- 2025 / Iterative neural networks for improving memory capacity. X Chen, D Lin, Z Li, W Li. Neural Networks 182, 106936, 2025
- 2024 / The hidden Markov model reveals the changes in brain dynamics among patients with end-stage renal disease under different dialysis methods. D Sun, W Li, H Yu, X Chen, C Zhang. Journal of Cyber-Physical-Social Intelligence 3 (1), 10-10, 2024
- 2024 / An Improved Algorithm for Spiking Neural Networks with Multi-Scale Attention Coding. S Chen, X Chen, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
- 2024 / Improved domain-adversarial neural networks with mixup and pseudo-labeling. X Xiao, X Chen, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
- 2024 / Dynamic Graph Contrastive Learning for Neuro Disease Identification using Resting-State fMRI. Y Jiang, W Li, X Chen. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
- 2024 / Synthesizing A β -PET based on multi-modal neuroimaging fusion for pathological diagnosis of Alzheimer ' s disease. Y Yang, B Li, W Li, H Chen, W Cao. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
- 2024 / Universum Inspired Graph Contrast Learning for Autism Spectrum Disorder Identification. H Qin, X Xu, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
- 2024 / Enhancing Functional Brain Network Construction for Neurological Disorder Diagnosis through Modularity-Driven Regularization. D Sun, X Chen, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024

- 2024 / Multi-view comparative self-supervised graph learning for neuro-disease diagnosis using functional brain networks. N Tian, X Xu, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
- 2024 / Topology-Enhanced Multi-Modal Residual Graph Convolutional Network for Autism Spectrum Disorder Identification. J Deng, L Guan, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
- 2024 / Efficient and accurate capsule networks with b-spline-based activation functions. L Mou, X Xiao, W Cao, W Li, X Chen. 2024 International Conference on New Trends in Computational Intelligence ... , 2024
- 2024 / The characteristic patterns of individual brain susceptibility networks underlie Alzheimer ' s disease and white matter hyperintensity-related cognitive impairment. H Chen, J Xu, W Li, Z Hu, Z Ke, R Qin, Y Xu. Translational Psychiatry 14 (1), 177, 2024
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- 2024 / Towards an accurate autism spectrum disorder diagnosis: multiple connectome views from fMRI data. J Yang, X Xu, M Sun, Y Ruan, C Sun, W Li, X Gao. Cerebral Cortex 34 (1), bhad477, 2024
- 2024 / Effects of individual metabolic brain network changes co-affected by T2DM and aging on the probabilities of T2DM: protective and risk factors. YL Li, JJ Wu, WK Li, X Gao, D Wei, X Xue, XY Hua, MX Zheng, JG Xu. Cerebral Cortex 34 (1), bhad439, 2024
- 2024 / Topological properties analysis and identification of mild cognitive impairment based on individual morphological brain network connectome. X Xu, P Chen, W Li, Y Xiang, Z Xie, Q Yu, Y Tang, P Wang. Cerebral Cortex 34 (1), bhad450, 2024
- 2023 / Wavelet transform-based frequency self-adaptive model for functional brain network. Y Ding, X Xu, L Peng, L Zhang, W Li, W Cao, X Gao. Cerebral Cortex 33 (22), 11181-11194, 2023
- 2023 / PageRank algorithm based on dynamic damping factor. Z HaoLin, H Jin, L WeiKai. 2023 International Conference on Cyber-Physical Social Intelligence (ICCSI ... , 2023
- 2023 / An improved AdaBoost method in imbalanced data Learning. T Li, X Chen, W Li. 2023 International Conference on Cyber-Physical Social Intelligence (ICCSI ... , 2023
- 2023 / Minimum error entropy high-order extended Kalman filter. Z Cheng, X Chen, H Li, W Li, D Lin. 2023 International Conference on Cyber-Physical Social Intelligence (ICCSI ... , 2023
- 2023 / New insights into the disorder of brain connectivity in schizophrenia. W Li, MM Al-Amin, A Nair. Frontiers in Neuroimaging 2, 1266695, 2023
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- 2022 / Partial Domain Adaptation without Domain Alignment. W Li, S Chen. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022
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- 2020 / Effective and accurate diagnosis of subjective cognitive decline based on functional connection and graph theory view. X Xu, W Li, M Tao, Z Xie, X Gao, L Yue, P Wang. *Frontiers in neuroscience* 14, 577887, 2020
- 2020 / Functional connectivity network estimation with an inter-similarity prior for mild cognitive impairment classification. W Li, X Xu, W Jiang, P Wang, X Gao. *Aging (Albany NY)* 12 (17), 17328, 2020
- 2020 / The effect of jet lag on the human brain: A neuroimaging study. F Zhang, W Li, H Li, S Gao, JA Sweeney, Z Jia, Q Gong. *Human brain mapping* 41 (9), 2281-2291, 2020
- 2020 / Group similarity constraint functional brain network estimation for mild cognitive impairment classification. X Gao, X Xu, X Hua, P Wang, W Li, R Li. *Frontiers in Neuroscience* 14, 165, 2020
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- 2019 / The altered reconfiguration pattern of brain modular architecture regulates cognitive function in cerebral small vessel disease. R Liu, H Chen, R Qin, Y Gu, X Chen, J Zou, YC Jiang, W Li, F Bai, *Frontiers in neurology* 10, 324, 2019
- 2019 / Functional brain network estimation with time series self-scrubbing. W Li, L Qiao, L Zhang, Z Wang, D Shen. *IEEE journal of biomedical and health informatics* 23 (6), 2494-2504, 2019
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- 2018 / 一种自适应分数阶偏微分图像增强模型.. 李伟凯, 王政霞, 蒋伟. *Computer Engineering & Science/Jisuanji Gongcheng yu Kexue* 40 (4), 699, 2018
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- 未标年份 / Kola: Carefully benchmarking world knowledge of large language models, 2024. J Yu, X Wang, S Tu, S Cao, D Zhang-Li, X Lv, H Peng, Z Yao, X Zhang, URL <https://arxiv.org/abs/2306.09296>
- 未标年份 / An Improved Efficient Algorithm for Spiking Neural Network Through Dataset Distillation. S Chen, X Chen, W Cao, W Li. Available at SSRN 4836784

- 未标年份 / Graph Adversarial Discriminative Domain Adaptation with Category-Level Alignment. X Xiao, X Chen, W Li, Z Li, W Cao, S Zhang. Available at SSRN 4898772

个人荣誉

- 2026 / CPSI 最佳论文奖
- 2025 / CPSI 最佳论文奖
- 2025 / 中国机器人及人工智能大赛优秀指导教师
- 2023 / ICCSI 最佳论文入围奖
- 2022 / 博士研究生国家奖学金

指导学生获奖

- 2026 / 崔靖轩等获 CPSI 2026 最佳论文奖
- 2025 / 中国机器人及人工智能大赛全国总决赛一等奖
- 2025 / 中国机器人及人工智能大赛重庆选拔赛一等奖
- 2025 / 全国数字艺术设计大赛全国三等奖
- 2025 / 美国大学生数学建模竞赛 H 奖、M 奖
- 2025 / 互联网+ 校赛一等奖
- 2025 / iCAN 校赛二等奖
- 2024 / 美国大学生数学建模竞赛 H 奖、M 奖
- 2024 / 挑战杯校赛一等奖、三等奖
- 2024 / 机器人竞赛重庆市二等奖
- 2024 / 校级创业大赛二等奖
- 2024 / 睿抗机器人开发者大赛一等奖
- 2023 / 全国大学生数学建模竞赛重庆赛区一等奖及二等奖
- 2023 / 中国软件杯西部赛区三等奖